

## KITEWRIGHT CORE · CALCULATION

# Kitewright Core sizing calculations

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## DOCUMENT CONTROL

|             |                                              |         |                |
|-------------|----------------------------------------------|---------|----------------|
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| Author      | Amish Chadha                                 | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept                |         |                |
| Repository  | github.com/BoujeeEnjinia1701/kitewright-core |         |                |
| Source      | docs/04-calcs/01-sizing.md @ uncommitted     |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                                                                                                               |
|---------|------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.2     | 2026-10-03 | Amish Chadha | Decision 33B carried out (KWC-DDR-003): carbon plate, pocketed rail bars, lighter pin blocks, corner spacers and lid, power leads moved to the frames; R3, R8, R9 and R11 recomputed; R9 now met on paper at 0.99 kg |
| 0.1     | 2026-10-03 | Amish Chadha | First issue for TRL 3 on the constructable design (KWC-DDR-002): mount strength, payload swap, payload power, heat, altitude, hover-time model, common core, mass and cost                                           |

## Kitewright Core sizing calculations

On paper the constructable core, lightened under Amish's decision 33B of 2026-10-03 ("i agree with all the 46 recommendations you provided. please proceed."), meets all eleven requirements, with one new point on R8 set out for Amish (open decision O2). The plain rail and two locking pins carry the 147 N vertical design load with factors of 8 or more on every part, a gloved payload swap takes about 52 s with no tools, 100 W reaches the payload at no more than 5.6 A, and the hottest avionics part stays near 75 °C at 45 °C ambient. The hover-time model explains the scaffold's 47 % figure: at 5,000 m and -20 °C an aircraft keeps 77.5 % of its sea-level hover time from air density alone, and unheated packs that give 60 % of their capacity bring it to 46.5 %; ColdCell packs kept warm raise it to 70.0 %. The core's estimated mass is 0.99 kg, 10 g inside R9's 1.0 kg, after the carbon plate, the pocketed rail bars and the move of the power leads to the frames. The estimated cost of USD 3,946.80, with ground station and spares, is USD 1,053.20 under the USD 5,000 value-engineering target. Every number here is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [A3], is the line of that script's output that carries it.

**Safety:** These are first-principles estimates for a paper proof of concept. They do not show that the payload mount, the lithium packs, the power wiring or the aircraft is safe. The mount must be proof-loaded and the locking pins pull-tested, and the power board must be checked on a current-limited bench supply, before any pack is connected or any payload is flown. See KWC-PRC-001, Safety.

### ■ Scope and method

The note checks every requirement in KWC-REQ-001 v0.2 against the design in KWC-PRC-001 v0.2 and the parametric model `cad/src/model.py`. The script imports the model's parameters and part volumes, so the plate, rails, shoe, pin blocks and lid used here are the ones in the STEP file, drawing KWC-DWG-001 and the build plan. It also reads `bom/bom.csv` and `budget_usd` in `project.yaml`. Run it from the repo root with `python docs/04-calcs/sizing.py`; it also writes `docs/04-calcs/results.csv`.

### ■ Assumptions

*Table 1. Main assumptions.*

| Area          | Assumption                                                                                                                                                                                                                                                                        | Basis                                                        |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Rated payload | 5 kg on the mount, one payload class for version 1 (KWC-DDR-001, D4)                                                                                                                                                                                                              | Kitewright Lift's sea-level payload                          |
| Load cases    | Static 3 x 5 kg; drop 1.5 x 5 kg at 2 g (both 147 N, R3); fore-aft 1.5 x 5 kg at 3 g (221 N) on one pin alone; side 1.5 x 5 kg at 1 g with the payload's centre of mass 80 mm below the shoe                                                                                      | R3 and screening values                                      |
| Materials     | 6061-T6 yield 276 MPa; 2 mm woven carbon plate: design allowable 250 MPa in bending and 200 MPa in bearing (typical strength 500 to 600 MPa, halved for holes, cold and moisture); hardened plunger pin 400 MPa in shear (conservative); M4 A2-70 screws at 0.8 x proof (3,161 N) | Handbook values; maker data for the plate class              |
| Swap          | Seven hand actions totalling 35 s bare-handed; gloves multiply by 1.5                                                                                                                                                                                                             | Estimate                                                     |
| Bus           | 18 V (6S Li-ion at 3.0 V a cell) to 60 V; 100 W payload; pigtail 0.3 m of 20 AWG, two pins each way; payload switch limit 8 A                                                                                                                                                     | KWC-DDR-001, D3                                              |
| Heat          | Lid surface 0.0477 m <sup>2</sup> (top and sides); 10 W/m <sup>2</sup> K in still air, 25 W/m <sup>2</sup> K in hover; 10 K allowance for sun on the frame body around the lid; 10 K from lid air to the hottest part; parts rated -40 to +85 °C                                  | Handbook ranges; maker ratings of the part classes           |
| Dissipation   | Flight controller 2.5 W, telemetry 1 W idle and 2 W in flight, receiver 0.3 W, 5.3 V supplies 1.1 W, payload switch 0.47 W, bus conduction 4 W at 100 A through 0.4 mΩ                                                                                                            | Estimates; 100 A is Kitewright Lift's hover estimate at 44 V |
| Atmosphere    | International Standard Atmosphere                                                                                                                                                                                                                                                 | ISA                                                          |
| Hover model   | Hover power proportional to 1/V(density) for the same mass and rotors; unheated Li-ion gives 60 % of capacity at -20 °C; ColdCell packs give 95 % and spend 5 % on their heater                                                                                                   | Momentum theory; typical cell data                           |
| Mass          | Made parts from model volumes (aluminium 2.70, carbon plate 1.55, ASA 1.07, G10 1.85, UHMW-PE 0.94 g/cm <sup>3</sup> ); bought parts from catalogue class values (Table 8)                                                                                                        | Estimates, to confirm when bought                            |

## ■ A. Mount strength (R3)

The vertical design load is 147 N in both R3 cases [A1]. The shoe overlaps each lip by 19 mm along 184 mm; the lip bends about the spacer bar's inner edge with a 10.5 mm lever, giving 2.8 MPa, a factor of 99 on yield [A2]. The lightening pockets under the lips leave a 1.2 mm floor that stops 2 mm short of the spacer bar; at the pocket's edge, taking the pocket along the whole shoe length, the lip reaches 14.2 MPa, a factor of 19 [A2]. The ten M4 screws that hold the lips see at most 46 N each with prying, a factor of 69 [A3]. Between the frame bolts, 220 mm apart, the carbon plate's edge strip, the windowed spacer bar and the pocketed lip, taken as three separate beams, reach 30 MPa, a factor of 8.3 on the carbon plate's 250 MPa allowable [A4]. With one pin alone holding the 221 N fore-aft case, its block's two M4 screws bear on the 2 mm carbon at 13.8 MPa, a factor of 14 [A8].

Fore and aft, the shoe is held by the front stops one way and by the locking pins both ways. The 221 N fore-aft case on ONE pin gives 11.2 MPa of shear in the 5 mm pin and 8.8 MPa of bearing in the shoe [A5]; forward load into the two front stops gives 1.2 MPa [A6]. A 1 g side load with the payload's centre of mass 80 mm down puts 54 N on each lip [A7]. Every factor is still large after decision 33B's lightening, because the parts were sized for making and handling, not for strength. That margin is why two independent pins cost little: either pin alone holds the payload.

The carbon plate needs two cautions that the numbers do not show: aluminium touching it is anodised so the pair cannot corrode in wet snow, and the M4 nuts on it sit on washers and are tightened to about 2 N·m so they do not crush the laminate.

## ■ B. Payload swap (R2)

Table 2. Swap sequence [B1].

| Action                                            | Time, bare-handed (s) |
|---------------------------------------------------|-----------------------|
| Unplug the DS-014 plug                            | 6                     |
| Pull and twist both locking pins to their rest    | 5                     |
| Slide the payload out to the rear                 | 4                     |
| Slide the next payload in to the front stops      | 5                     |
| Twist both pins back; they snap into the shoe     | 3                     |
| Plug the DS-014 plug and close its latch          | 8                     |
| Check both pins show no red band; tug the payload | 4                     |
| Total                                             | 35                    |

With gloves the swap takes about 52 s and needs no tools. The margin to R2's figure is small, so a timed gloved trial at TRL 4 is the real check.

## ■ C. Payload power (R4)

Table 3. Payload current at the end of discharge [C1].

| Bus                                    | Voltage (V) | Current for 100 W (A) | Pigtail drop (mV) |
|----------------------------------------|-------------|-----------------------|-------------------|
| 6S Li-ion, 3.0 V a cell                | 18.0        | 5.56                  | 56                |
| 12S Li-ion                             | 36.0        | 2.78                  | 28                |
| 16S LiFePO <sub>4</sub> , 2.8 V a cell | 44.8        | 2.23                  | 22                |

The 8 A payload switch covers the worst case with a 1.44 x margin [C2]. The current each DS-014 battery pin may carry is to be confirmed from the published standard and the chosen connector; two pins each way at 3 A would cover the 6S case.

## ■ D. Heat (R5)

The lid encloses 4.9 W on the ground and 10.4 W in Kitewright Lift's hover, where 4 W comes from the 100 A bus current [D1]. On a 45 °C day, with 10 K allowed for sun on the frame body, the lid air reaches 65.3 °C on the ground and 63.7 °C in hover, and the hottest part about 75 °C, 10 K inside the 85 °C rating [D2]. Cold-soaked to -20 °C, every part boots within its -40 °C rating, and once running the lid air settles near -10 °C [D3]. The optional companion computer sits outside the lid (KWC-DDR-001, D8), so its 5 W does not count here.

## ■ E. Altitude (R6)

Table 4. International Standard Atmosphere [E1].

| Altitude (m) | Temperature (°C) | Pressure (kPa) | Density (kg/m³) | Share of sea-level density |
|--------------|------------------|----------------|-----------------|----------------------------|
| 0            | 15.0             | 101.3          | 1.225           | 100 %                      |
| 3,000        | -4.5             | 70.1           | 0.909           | 74 %                       |
| 5,000        | -17.5            | 54.0           | 0.736           | 60 %                       |
| 6,000        | -24.0            | 47.2           | 0.660           | 54 %                       |

At 5,000 m and -20 °C the density is 0.743 kg/m³, and the barometer must read down to 54 kPa; flight-controller barometers of this class read 30 to 110 kPa [E2]. The core's share of R6 is sensors in range and failsafes set for each band (Table 6). Whether a frame has the thrust to stay in control is a frame property, checked in Kitewright Lift and Kitewright Range with AltiRig data.

Table 6. Altitude parameter sets: what each band changes (released PX4, parameters only).

| Setting                        | Sea level band (0 to 1,500 m)           | 3,000 m band (1,500 to 4,000 m) | 5,000 m band (4,000 to 6,000 m)          |
|--------------------------------|-----------------------------------------|---------------------------------|------------------------------------------|
| Hover thrust estimate          | From the frame's sea-level hover test   | Sea-level value x 1.16          | Sea-level value x 1.29                   |
| Low-battery warning and return | 30 % and 20 %                           | 35 % and 25 %                   | 40 % and 30 %                            |
| Return altitude above launch   | 30 m                                    | 50 m                            | 80 m (terrain)                           |
| Link-loss action               | Return after 5 s                        | Return after 5 s                | Return after 3 s                         |
| Geofence                       | Set per site, 1,000 m radius by default | As sea level                    | As sea level, 120 m ceiling above launch |
| Pack temperature before arming | ColdCell reports at least 10 °C         | As sea level                    | As sea level                             |

The hover-thrust factors are  $1/\sqrt{\text{density share}}$  from Table 4; the battery thresholds are deliberately higher at altitude because a return through thin air takes more energy. All values are starting points to be tuned on AltiRig and in flight.

## ■ F. Hover-time model (R7)

For the same aircraft and mass, hover power scales with  $1/\sqrt{\text{density}}$ , so hover time at 5,000 m is 0.775 of sea level from air alone [F1]. Unheated packs that give 60 % of their capacity at -20 °C bring that to 46.5 %, which is the scaffold's 47 % estimate [F2]. ColdCell packs kept warm, giving 95 % of their capacity and spending 5 % on their heater, bring it to 70.0 % [F3]. The model is:

hover time = usable pack energy x (1 - heater share) x efficiency / hover power, with hover power =  $(m g)^{1.5} / (\text{figure of merit} \times \sqrt{2 \times \text{density} \times \text{rotor disc area}})$ .

It leaves out the change in motor and propeller efficiency at the higher rotor speeds of thin air; AltiRig measures that. R7 asks that the model predict measured hover time within 15 %, which only TRL 4 flights can show.

## ■ G. Common core (R8, R10, R1)

Table 7. Outputs needed on the flight controller [G1].

| Frame                                           | Outputs needed | Available |
|-------------------------------------------------|----------------|-----------|
| Lift, eight motors, plus three payload channels | 11             | 16        |
| Lift, six motors, plus three payload channels   | 9              | 16        |
| Range, four lift motors, pusher and five servos | 10             | 16        |

One core, one frame pattern and one payload interface serve both frames with parameter changes only (R8). Released PX4 (v1.15 or later) runs multirotor and standard VTOL frames, DroneCAN battery reporting and MAVLink payload protocols without source changes (R10). The DS-014 pigtail carries the battery, Ethernet, CAN, UART and trigger lines named in the standard (R1); the exact pinout and licence terms are confirmed from the published standard before the connector is bought.

## ■ H. Mass (R9)

Table 8. Core mass after decision 33B [H1, H2].

| Part                                                                                 | Mass (g) | Source          |
|--------------------------------------------------------------------------------------|----------|-----------------|
| Core plate, 2 mm carbon                                                              | 109      | Model volume    |
| Rail spacer bars (windowed), lips (pocketed) and wear tape                           | 118      | Model volume    |
| Front stops, pin blocks (12 mm), jam nuts, corner spacers (12 mm)                    | 77       | Model volume    |
| Lid, 1.5 mm walls                                                                    | 90       | Model volume    |
| Damping plate and strain-relief bar                                                  | 30       | Model volume    |
| Flight controller                                                                    | 100      | Catalogue class |
| Power board, power modules                                                           | 105      | Catalogue class |
| Radio, receiver, antennas and bulkheads                                              | 91       | Catalogue class |
| GNSS receiver and mast                                                               | 45       | Catalogue class |
| Locking pins (2)                                                                     | 50       | Catalogue class |
| DS-014 pigtail, harness, switch, grommets, gasket                                    | 110      | Catalogue class |
| Dampers, standoffs, countersunk fixings, strain-relief standoffs and ties, fasteners | 64       | Catalogue class |
| Total (rows rounded)                                                                 | 989      |                 |

The core weighs about 0.99 kg without the payload shoe (310 g, counted with each payload) and without the optional companion computer (60 g, outside the lid) [H2]. That is 10 g inside R9's 1.0 kg [H3], and 284 g lighter than the 1.27 kg core drawn before decision 33B [H4]. The saving comes from the 2 mm carbon fibre plate (81 g), windows in the spacer bars and pockets under the lips (44 g), 12 mm pin blocks and 12 mm corner spacers (20 g), 1.5 mm lid walls (24 g), and the four 8 AWG power leads with their AS150 plugs (124 g), which now belong to each frame's harness; the strain-relief bar and the countersunk fixings that replace the clinch nuts add back about 9 g. The frames carry the leads instead, so the aircraft's mass is unchanged overall [H5]. The 10 g margin is thin against catalogue-class masses for the bought parts, so weighing the bought parts when they arrive (register, T9) decides it.

## ■ I. Cost (R11)

The bill of materials comes to USD 1,766.80 for the airborne core, USD 1,620.00 for the ground station and USD 560.00 for spares [I1]. Decision 33B adds about USD 58 for the lighter structure (carbon plate USD 40, milling and anodising the rail bars USD 20, USD 1 less filament), USD 8 for the strain-relief bar and USD 0.80 less in fixings, and removes the USD 60 of leads and plugs that the frames now supply: USD 6.20 more in all. Value-engineering target: USD 5,000. Estimated cost of the constructable design: USD 3,946.80 (USD 1,053.20 under the target) [I2].

## ■ L. Results against every requirement

*Table 9. Results (also in `docs/04-calcs/results.csv`).*

| ID  | Result                                                                                                                                                | Status                                                                           |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| R1  | DS-014 40-pin connector on a 0.3 m pigtail; signals from the flight controller and bus                                                                | Met on paper (licence and pinout to confirm)                                     |
| R2  | About 52 s with gloves, no tools; pin state shown by a red band                                                                                       | Met on paper (estimate)                                                          |
| R3  | Vertical 147 N: lips 19 x at the pockets, screws 69 x, carbon side beam 8.3 x; fore-aft 221 N on one pin 36 x, carbon bearing 14 x                    | Met on paper                                                                     |
| R4  | 100 W needs at most 5.6 A (6S at 18 V); switch limit 8 A                                                                                              | Met on paper (pin rating to confirm)                                             |
| R5  | Hottest part about 75 °C at 45 °C ambient; cold boot at -20 °C within the -40 °C rating                                                               | Met on paper                                                                     |
| R6  | Sensors in range at 54 kPa; failsafes set per altitude band                                                                                           | Met on paper for the core; flight check needs a frame                            |
| R7  | Model published: 47 % (cold packs) and 70 % (ColdCell) of sea-level hover time at 5,000 m and -20 °C                                                  | Model ready; accuracy shown only at TRL 4                                        |
| R8  | One core and one interface; Lift needs 11 of 16 outputs, Range 10; moving the core between frames now also changes the frame's soldered power harness | Met by design for firmware and fixing; harness change open (O2, docs/REVIEW.md ) |
| R9  | 0.99 kg estimated (decision 33B)                                                                                                                      | Met on paper, 10 g margin; bought-part masses to confirm                         |
| R10 | Released PX4, parameters and stock modules only; ColdCell reports over DroneCAN                                                                       | Met by design                                                                    |
| R11 | USD 3,946.80 including ground station and spares                                                                                                      | Under the value-engineering target by USD 1,053.20                               |

## ■ Checks against earlier figures

- The scaffold's estimate of 47 % hover time at 5,000 m and -20 °C is reproduced (46.5 %) and traced to unheated packs; with ColdCell the model gives 70 %.
- R9 was written before the core had a constructable structure; the constructable design as first drawn missed it at 1.27 kg (version 0.1). Under decision 33B it meets it at 0.99 kg, within 1 % of the 0.99 kg estimated for option B in version 0.1.